Regularization for Cholesky Decomposition in Massive MIMO Detection

Alexander Osinsky, Roman Bychkov, Mikhail Trefilov, Vladimir Lyashev, and Andrey Ivanov

***Abstract*—In this letter, we propose a new regularization approach to low-precision Cholesky decomposition. This decomposition is employed to inverse an ill-conditioned matrix in the linear detector of the Multi-User Massive Multiple Input, Multiple Output (MU-MIMO) receivers. The proposed method was implemented in half-precision format and tested in realistic scenarios generated by the QuaDRiGa 2.0 channel. Simulations show that our approach demonstrates performance close to the single precision implementation. In high interference scenarios, it outperforms state-of-the-art alternatives, such as regularization based on weighted noise power.**

***Index Terms*—Cholesky decomposition, massive MIMO.**

I. Introduction

TO MEET the capacity requirements of the upgraded Mobile Broadband, massive multiple-input multiple-output (MIMO) was adopted as the primary technology in the fifth-generation (5G, [1]) of wireless networks. While the number of antennas in a standard 4G MIMO is $N_{RX}^{4G} = 8$, it begins at $N_{RX}^{5G} = 64$ in massive MIMO [1]. As a result, the excessive complexity of the MIMO receiver [2] unit stands as a major issue. The complexity of the linear Minimum Mean Squared Error (MMSE) MIMO detector in 5G is at least $[N_{RX}^{5G}/N_{RX}^{4G}]^2_{min} = 64$ times greater than in 4G. Linear MMSE has the lowest complexity, compared to QR-based [3] or maximum likelihood [4] detectors, which is especially important when the number of antennas is large. It is also known to be optimal for Gaussian noise and data [5].

In real-world applications, the number of users and the number of propagation channel taps are limited, so there is a limited set of directions-of-arrivals (DoA) at the receiver. As a result, the antenna signal transformation to a reduced dimensionality beamspace [6] is widely employed to address issues of high complexity in a Massive MIMO receiver. Thus, after transformation to the beamspace domain, the received Orthogonal Frequency Division Multiplexing (OFDM) signal $Y \in \mathbb{C}^{N_{beam}}$ on a single subcarrier [2] is given by:

$$Y = HX + E + V, \tag{1}$$

where $N_{beam} < N_{RX}$ is the beamspace size, $H \in \mathbb{C}^{N_{beam} \times N_{TX}}$ is the channel matrix, $X \in \mathbb{C}^{N_{TX}}$ is the data symbols, transmitted by N_{TX} antennas for all users, $E \in \mathbb{C}^{N_{beam} \times N_{SC}}$ is the additive white Gaussian noise (AWGN) with power σ^2 per beam, $V \in \mathbb{C}^{N_{beam}}$ is interference.

A. MMSE Detection

The MMSE detector can be written as follows [2]:

$$\hat{X} = \left(H^H R_{UU}^{-1} H + I\right)^{-1} H^H R_{UU}^{-1} Y, \tag{2}$$

where $R_{UU} \in \mathbb{C}^{N_{beam} \times N_{beam}}$ is the interference+noise correlation matrix. The most resource-intensive parts of the algorithm are the computation of R_{UU}^{-1} and its multiplication by H, both of which depend on the number of beams N_{beam}.

Let us denote $U_i = Y - H^{\text{est}} X$, $U_i \in \mathbb{C}^{N_{beam}}$ the noise+interference estimate for any particular subcarrier i, where H^{est} is the channel estimate. Usually, both matrices H and R_{UU} are estimated over pilot symbols [2], [7]. Combining all subcarriers, we get the matrix $U \in \mathbb{C}^{N_{beam} \times N_{SC}}$, which is then used to estimate R_{UU} [2]:

$$R_{UU}^{\text{est}} = \frac{1}{N_{SC}} \sum_{i=1}^{N_{SC}} U_i U_i^H = \frac{1}{N_{SC}} U U^H, \tag{3}$$

where N_{SC} is the number of allocated subcarriers.

While R_{UU} needs to be calculated and inverted only once per frame, it still can be the most expensive part of the algorithm, since R_{UU} matrix has the size much larger than H for $N_{beam} \gg N_{TX}$. As a result, one is forced to use the fastest possible algorithm to invert it, which is given by Cholesky factorization $R_{UU} = LL^H$ [8].

B. Problem Definition

The problem with using Cholesky factorization arises when one tries to reduce the complexity of the detector even further, by using low-precision arithmetic (i.e., 16-bit half-precision instead of 32-bit single precision) [9]. In this case, the accuracy of the detector significantly depends on the condition number of R_{UU} [10], which is closely related to the interference-to-noise ratio. When the interference is high, numerical errors of the matrix inverse become huge, and Cholesky decomposition can even become unstable. As a result, the regularization

$$\hat{R}_{UU} = R_{UU}^{\text{est}} + \alpha I \tag{4}$$

is usually used to make Cholesky decomposition stable again [11], where α is the regularization parameter, and I is the identity matrix. Unfortunately, large values of α bring significant errors to $\hat{R}_{UU}^{-1}$. Depending on the value of α one can also see a degradation of performance when the condition number is not large. Consequently, there is an open problem of how one can improve the regularization to guarantee the stability of Cholesky decomposition for low-precision arithmetic, make the errors introduced by regularization smaller, and guarantee the best performance. We solve these problems here.

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C. Our Contribution

In this letter, we propose new regularization methods to increase the accuracy of matrix inversion over Cholesky decomposition in half precision arithmetic for application to the MMSE detector in Massive MIMO receiver. Simulation results with the Quadriga 2.0 [12] channel are provided.

II. Regularization for Cholesky

A. Optimal Regularization

Let us explore the standard regularization methods. First, we need to note that even when the condition number of R_{UU} is small, optimal regularization is not zero. This comes from the fact that R_{UU} the matrix in the detector (2) is not exact and instead given by (3), which can make its lowest singular values significantly lower than the noise power.

Since interference is a low-dimensional subspace, it does not significantly affect the lowest eigenvalues of R_{UU}, which, under the assumption of white Gaussian noise, are then not far from eigenvalues of the complex Wishart ensemble [13]:

$$\begin{aligned} R_{UU}^{\text{est}} &= \frac{1}{N_{SC}} VV^H + R_{UU}^{\text{noise}}, \\ R_{UU}^{\text{noise}} &= \frac{1}{N_{SC}} EE^H, \quad E \in \mathbb{C}^{N_{beam} \times N_{SC}}, \end{aligned} \tag{5}$$

where E consists of independent normal random variables with variance $\mathbb{E}|E_{ij}|^2 = \sigma^2$. In the limit, when N_{beam} and N_{SC} both converge to infinity, while $\frac{N_{beam}}{N_{SC}} \to \lambda$, eigenvalues of R_{UU}^{noise} are distributed according to Marchenko-Pastur distribution (according to its definition), with the following minimum eigenvalue:

$$\lambda_{\min}\left(R_{UU}^{\text{noise}}\right) \to \left(1 - \sqrt{\lambda}\right)^2 \sigma^2 = \left(1 - \sqrt{\frac{N_{beam}}{N_{SC}}}\right)^2 \sigma^2. \tag{6}$$

Thus, in the case $N_{SC} \gg N_{beam}$ we can assume that the lowest eigenvalue is close to $(1 - \sqrt{\frac{N_{beam}}{N_{SC}}})^2\sigma^2$. Strict proof of the convergence for the minimum eigenvalue and better estimates for it can be found in [13].

Equation (6) suggests using the regularization

$$\hat{R}_{UU} = R_{UU}^{\text{est}} + \alpha I \tag{7}$$

with α close to

$$\alpha_{\text{opt}} \approx \left(1 - \left(1 - \sqrt{\frac{N_{beam}}{N_{SC}}}\right)^2\right)\sigma^2, \tag{8}$$

so that the lowest eigenvalue of R_{UU} after regularization will be close to $\lambda_{\min} + \alpha \approx \sigma^2$. As a result, the estimated noise power will coincide with the real noise power. For $N_{SC} < N_{beam}$ the lowest eigenvalue of R_{UU}^{est} is zero, so the optimal regularization is $\alpha \approx \sigma^2$. While this regularization also increases large eigenvalues, we can neglect the error for them, since we assumed that interference power is much larger.

B. Stable Regularization

The problem with standard regularization appears when numerical errors are introduced. Let's say that we store the values of R_{UU}^{est} up to relative precision ε. Then, to guarantee the stability of Cholesky, it is required that the absolute error (which is of the order $\varepsilon \max_i R_{UU,ii}^{\text{est}}$) cannot make the R_{UU} matrix degenerate. As shown in [14], the error usually propagates linearly, so when the regularization

$$\alpha = c \max_i R_{UU,ii}^{\text{est}} \gtrsim \varepsilon N_{beam} \max_i R_{UU,ii}^{\text{est}} \tag{9}$$

is introduced, it should compensate for any numerical errors.

The problem with this kind of regularization is that if one tries to fit the coefficient c, the result is going to be far from optimal when the condition number of R_{UU} is small and Cholesky is always stable. One can, obviously, try to switch between regularizations (9) and (8) depending on the interference-to-noise ratio, but there is a better approach.

C. Combined Regularization

We suggest combining (9) and (8) by observing that whenever (8) is larger than (9), it guarantees the stability of Cholesky decomposition. Optimal regularization should be always preferred in this scenario, which leads to the combined regularization formula

$$\alpha = \max\left(\alpha_1\sigma^2, \alpha_2 \max_i R_{UU,ii}^{\text{est}}\right), \tag{10}$$

where α_1 and α_2 depend on the particular values of N_{beam}, N_{SC}, N_P and ε. Here $\alpha_1\sigma^2$ is the optimal regularization, which would always be preferred if we did not have numerical errors, which could force us to use α_2 instead. While it is possible to compute rough estimates for α_1 and α_2 (like (8)), optimal values can be learned and set in advance for each particular receiver.

Note, that if the second regularization (based on α_2) is chosen, a significant error is introduced. On the other hand, if we do not introduce this regularization, we cannot guarantee the convergence of Cholesky decomposition. Even if one tries to replace Cholesky with, for example, LDL^H decomposition, modified Cholesky [15] or some other alternative, one still requires a similar value of α to guarantee non-degeneracy. And even if one is lucky to get a non-degenerate matrix, the error of $\hat{R}_{UU}^{-1}$ is still going to be large.

D. Scaling R_{UU}

To reduce the error, coming from increased regularization, one can try to scale R_{UU} appropriately. Assume that interference forms a low-dimensional interference subspace, defined by the matrix $V_{\text{sub}} \in \mathbb{C}^{N_{beam} \times N_{int}}$ (where we explicitly reduced the number of columns), such that $VV^H = V_{\text{sub}} V_{\text{sub}}^H$. Then exact noise+interference correlation matrix R_{UU}^{ideal} is

$$R_{UU}^{\text{ideal}} = V_{\text{sub}} V_{\text{sub}}^H + \sigma^2 I. \tag{11}$$

However, in practice, one uses instead

$$\hat{R}_{UU} = VV^H + EE^H + \alpha I \approx V_{\text{sub}} V_{\text{sub}}^H + \alpha I, \tag{12}$$

since $\mathbb{E}(EE^H) = \sigma^2 I \ll \alpha I$ if $\alpha \gg \sigma^2$ due to α_2 being chosen in combined regularization (10).

Let $V_{\text{sub}} = QR$, $Q \in \mathbb{C}^{N_{beam} \times N_{int}}$, $Q^H Q = I$, $R \in \mathbb{C}^{N_{int} \times N_{int}}$. Assuming all eigenvalues of R are much larger

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than regularization, after inversion we get almost zero eigenvalues in the subspace, corresponding to Q and σ^{-2} (or α^{-1}) eigenvalues in the orthogonal subspace $I - QQ^H$, so

$$\left(R_{UU}^{\text{ideal}}\right)^{-1} \approx \left(I - QQ^H\right)\sigma^{-2},$$
$$\hat{R}_{UU}^{-1} \approx \left(I - QQ^H\right)\alpha^{-1}. \tag{13}$$

So, if $\alpha > \sigma^2$, it is reasonable to scale the R_{UU} matrix by a factor σ^2/α, which leads to the detector of the form

$$\hat{X} = \left(H^H \frac{\alpha}{\sigma^2} R_{UU}^{-1} H + I\right)^{-1} H^H \frac{\alpha}{\sigma^2} R_{UU}^{-1} Y$$
$$= \left(H^H R_{UU}^{-1} H + \frac{\sigma^2}{\alpha} I\right)^{-1} H^H R_{UU}^{-1} Y. \tag{14}$$

While this scaling does introduce the appropriate noise power in the non-interference direction, the main problem with regularization is that the ratio between interference and noise power is estimated incorrectly, thus leading to interference not being dumped enough. Thus, as we shall see, this idea does not lead to significant performance improvement.

E. Scaling L

To be more precise with the scaling, we suggest, in addition to the previous subsection, introducing diagonal scale S:

$$\hat{R}_{UU}^{\text{scaled}} = LS^2L^H, \tag{15}$$

where $S \in \mathbb{C}^{N_{beam} \times N_{beam}}$ is a diagonal matrix and

$$LL^H = \hat{R}_{UU} \tag{16}$$

is the Cholesky decomposition of $\hat{R}_{UU}$.

As previously mentioned, the main problem is that the ratio between noise and interference power is not preserved. After Cholesky we get

$$L_{kk} = \sqrt{\tilde{R}_{UU,kk}^{\text{est}} + \alpha}, \tag{17}$$

where $\tilde{R}_{UU,kk}^{\text{est}}$ is the k-th diagonal element of R_{UU}^{est} after $k-1$ Cholesky steps, applied to $\hat{R}_{UU}$ [11]. Here regularisation α is given by (10). However, we would prefer to use optimal regularisation α_{opt} (8) instead and thus obtain

$$L_{kk}^{\text{opt}} = \sqrt{\tilde{R}_{UU,kk}^{\text{est}} + \alpha_{\text{opt}}} \tag{18}$$

with $\alpha_{\text{opt}} = \alpha_1\sigma^2$. Now, in (17) the ratio between regularization and the diagonal element is $\alpha/\tilde{R}_{UU,kk}^{\text{est}}$, while in (18) the ratio is $\alpha_{\text{opt}}/\tilde{R}_{UU,kk}^{\text{est}} = \alpha_1\sigma^2/\tilde{R}_{UU,kk}^{\text{est}}$. Note that we only care about the ratio between the diagonal element and its regularisation and not the regularisation itself, since we rescale R_{UU} afterward (14) by a factor $\frac{\sigma^2}{\alpha}$. Therefore, after rescale of L we should get L_{kk}^{opt} multiplied by a factor $\sqrt{\frac{\alpha}{\sigma^2}}$:

$$S_{kk}L_{kk} = \sqrt{\frac{\alpha}{\sigma^2}} L_{kk}^{\text{opt}} \approx \sqrt{\frac{\alpha}{\alpha_{\text{opt}}}} L_{kk}^{\text{opt}} = \sqrt{\frac{\alpha}{\alpha_1\sigma^2}\tilde{R}_{UU,kk}^{\text{est}} + \alpha}. \tag{19}$$

If $\alpha_{\text{opt}} = \alpha_1\sigma^2 \approx \sigma^2$ is not satisfied, we can use parameters p_0 and f_0 introduced below to compensate.

Let us write the scale in the following form:

$$S_{kk} = \sqrt{p^2 + (1-p^2)f^2}, \tag{20}$$

where f is the scale factor and $0 \le p \le 1$ is the part to be scaled as $p = \sqrt{p_0\alpha}/L_{kk}$, where p_0 is the parameter. Using $p_0 = 1$ and $f = \sqrt{\alpha/(\alpha_1\sigma^2)}$, we get

$$L'_{kk} = S_{kk}L_{kk} = \sqrt{\alpha + \left(L_{kk}^2 - \alpha\right)\frac{\alpha}{\sigma^2}}$$
$$= \sqrt{\alpha + \tilde{R}_{UU,kk}^{\text{est}}\frac{\alpha}{\alpha_1\sigma^2}}, \tag{21}$$

which agrees with the desired result (19).

To get more freedom, we also optimize the value of f. Instead of using $f = \sqrt{\alpha/(\alpha_1\sigma^2)}$, we use

$$f = 1 + f_0\left(\sqrt{\frac{\alpha}{\sigma^2}} - 1\right). \tag{22}$$

Thus, we have two parameters to optimize: p_0 and f_0.

F. Non-Identity Regularization

A much more promising idea is to use arbitrary diagonal regularization D. A similar idea was suggested in [10] based on the error analysis from [16]. The authors applied regularization αI to the symmetric positive definite matrix $A \in \mathbb{R}^{N\times N}$ as

$$\bar{A} = cI + \operatorname{diag}\left(A_{11}^{-1/2}, \ldots, A_{NN}^{-1/2}\right) A \operatorname{diag}\left(A_{11}^{-1/2}, \ldots, A_{NN}^{-1/2}\right), \tag{23}$$

scaled by its diagonal elements, and then scale $\bar{A}$ back:

$$\bar{A} = \bar{L}\bar{L}^H, \qquad L = \operatorname{diag}\left(A_{11}^{1/2}, \ldots, A_{NN}^{1/2}\right)\bar{L}. \tag{24}$$

Scaling back before Cholesky decomposition instead of scaling after it leads us to

$$A' = \operatorname{diag}\left(A_{11}^{1/2}, \ldots, A_{NN}^{1/2}\right)\bar{A}\operatorname{diag}\left(A_{11}^{1/2}, \ldots, A_{NN}^{1/2}\right)$$
$$= A + c\operatorname{diag}(A_{11}, \ldots, A_{NN}). \tag{25}$$

Thus, for $A = R_{UU}^{\text{est}}$ and $c = \alpha_2$ we suggest regularization:

$$\hat{R}_{UU} = R_{UU}^{\text{est}} + D, \tag{26}$$

where

$$D_{kk} = \max\left(\alpha_1\sigma^2, \alpha_2 R_{UU,kk}^{\text{est}}\right). \tag{27}$$

Equation (27) is the main difference in our non-identity regularization compared to the approach from [10] and [16]. Namely, like in combined regularization, optimal regularization is used when possible. When finding optimal α_2 in (27), one could get a larger value than in (10), but the fact that each diagonal element of D_{kk} is specific to the diagonal elements of R_{UU}^{est} leads to a lower total regularization. Therefore, relative regularization is much lower than in (10), where we applied regularization $\alpha_2 \max_i R_{UU,ii}^{\text{est}}$ to elements $R_{UU,kk}^{\text{est}} \ll R_{UU,ii}^{\text{est}}$, and this is where most of the error comes from.

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G. Scaling for Non-Identity Regularization

Now, if we want to scale $\hat{R}_{UU} = R_{UU}^{\text{est}} + D$, we cannot do it directly, since diagonal elements of D are significantly different, and the reasoning from Section II-D is no longer valid. As an alternative, we suggest again using (15).

Now we need to determine the values of S_{kk}. Observe that

$$L_{kk} = \sqrt{\tilde{R}_{UU,kk}^{\text{est}} + D_{kk}}, \tag{28}$$

where $\tilde{R}_{UU,kk}^{\text{est}}$ is the k-th diagonal element of R_{UU}^{est} after $k-1$ Cholesky steps. We again want to have L as if regularization was $\alpha_{\text{opt}} = \alpha_1\sigma^2$ instead of D_{kk} (see Section II-E), but now we do not use the R_{UU} scale (14) as the ratio D_{kk}/σ^2 is different for different diagonal elements.

Define the diagonal elements of S in the following form:

$$S_{kk} = \sqrt{p_k^2 f_k^2 + (1 - p_k^2)}, \tag{29}$$

where, again, f_k is the scale factor and

$$p_k = \min\left(p_0 \frac{\sqrt{D_{kk}}}{L_{kk}}, 1\right) \tag{30}$$

is the part of L_{kk} needed to be scaled, where we introduced the parameter p_0 and used the fact, that this part cannot be greater than the whole L_{kk} (thus, $p_k \leq 1$). If $p_0 = 1$ and $f_k = \sqrt{\alpha_1\sigma^2/D_{kk}}$, we get the value

$$\begin{aligned} L'_{kk} = S_{kk} L_{kk} &= \sqrt{f_k^2 D_{kk} + \left(L_{kk}^2 - D_{kk}\right)} \\ &= \sqrt{\alpha_1\sigma^2 + \tilde{R}_{UU,kk}^{\text{est}}} = L_{kk}^{\text{opt}} \end{aligned} \tag{31}$$

as if we used optimal regularization. To get more freedom, we also optimize the value of f. Instead of using $f_k = \sqrt{\alpha_1\sigma^2/D_{kk}}$, we use

$$f_k = 1/\left(1 + f_0\left(\frac{\sqrt{D_{kk}}}{\sigma} - 1\right)\right), \tag{32}$$

when $D_{kk} > \sigma^2$, otherwise $f_k = 1$. Thus, we again have two additional parameters: p_0 and f_0, which can be optimized.

III. Regularization Complexity

We should note that regularization $\hat{R}_{UU} = R_{UU}^{\text{est}} + D$ has a cost of $O(N_{beam})$ operations, whether elements of D are equal or not. Computation of the matrix D or the regularization parameter α also takes at most $O(N_{beam})$ operations for all the described methods (8), (9), (10), (27). This is much lower than the complexity of computing R_{UU}^{est} (3), equal to $O(N_{beam}^2 N_{SC})$, and the complexity of the Cholesky decomposition $O(N_{beam}^3)$.

The scale of L through multiplication by the diagonal matrix S in both Sections II-E and II-G requires $O(N_{beam}^2)$ arithmetic operations, which is also lower than $O(N_{beam}^3)$. Consequently, all the described regularization methods have a negligible effect on the total complexity of the detector. The parameters α_1, α_2, p_0 and f_0 are trained in advance based on the number of beams, subcarriers and numerical precision, which does not affect the total complexity.

TABLE I
Simulation Parameters

Parameter	Value	Parameter	Value
Carrier frequency	3.5GHz	**Modulation**	QPSK
Subcarrier spacing	30kHz	**N of scenarios**	140
N_{SC}	48	**Decoder**	LDPC (144,288)
N_{beam}	32	**RB number**	4
Noise seeds per scenario	32	**Channel model**	3GPP-3D, Berlin, Dresden
N_{RX}	64	**Channel type**	NLOS
N_{TX} **(2 per UE)**	2, 4, 8	**N of target UEs**	1, 2, 4
User speed	5 km/h	**N of interferers**	1, 2, 4, 8

TABLE II
Regularization Variants

Name	Regularization	Scale R_{UU}	Scale L
Optimal	Eq. (8)	No	No
Stable	Eq. (9)	No	No
Combined	Eq. (10)	No	No
Scaled R_{UU}	Eq. (10)	Yes	No
Scaled L	Eq. (10)	Yes	Yes
Non-identity	Eq. (27)	No	Yes

TABLE III
Trained Parameters

Name	α_1	α_2	f_0	p_0
Optimal	0.97	–	–	–
Stable	–	0.0019	–	–
Combined	0.95	0.0018	–	–
Scaled R_{UU}	1.85	0.0020	–	–
Scaled L	1.88	0.0022	0.66	0.92
Non-identity	2.28	0.0090	0.31	1.26

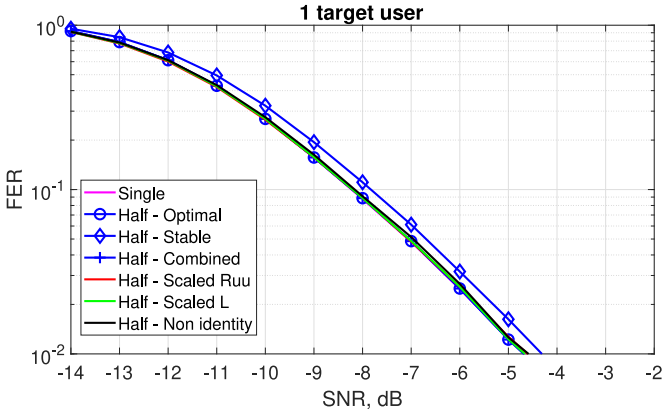


Fig. 1. 1 interferer with a power of 10dB above AWGN.

IV. Simulation Results

We employed the Quadriga 2.0 software [12] to generate a realistic radio channel for both target users and interferers, and the algorithm [17] to estimate the channel. For parameters training and simulations, we used scenarios described in Table I. The number of users is indicated above each figure. Based on the ideas from Section II, one can try multiple variants of the regularization, which we describe in Table II. To find realistic values, we fit the parameters α_1, α_2, f_0, and p_0 using a genetic algorithm (GA) to minimize the mean squared error (MSE) of the MIMO detector. Frame error rate (FER) dependencies on the signal-to-noise ratio (SNR) at receive antenna element (in dB) are presented in Figs. 1–4, where

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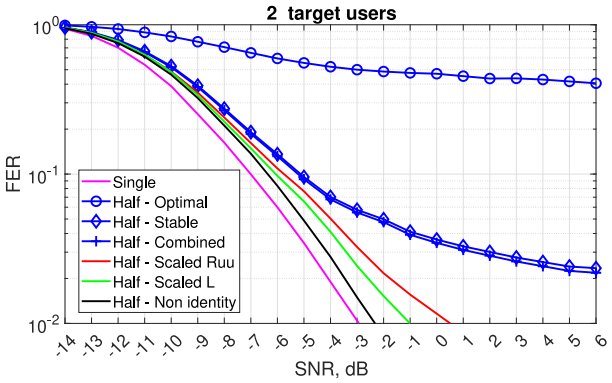


Fig. 2. 1 interferer with a power of 20dB above AWGN.

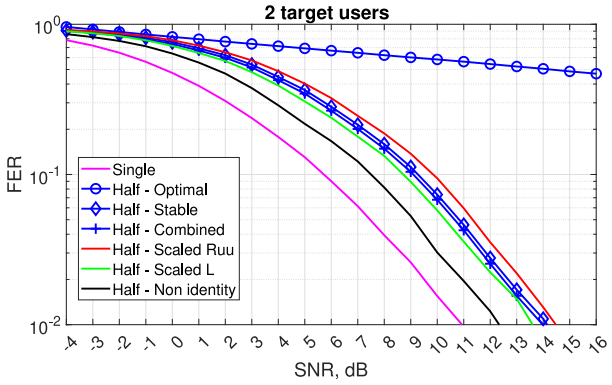


Fig. 3. 4 interferers, each has a power of 24dB above AWGN.

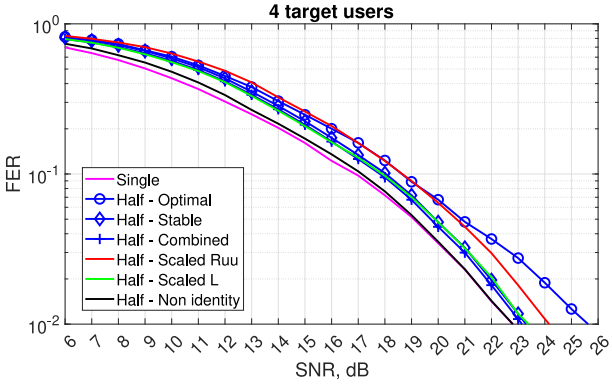


Fig. 4. 8 interferers, each has a power of 21dB above AWGN.

the MMSE detector is fully implemented in either half- or single-precision formats.

Note that we cannot additionally scale R_{UU} in the **Non-identity** regularization, as the scale depends on the regularization parameter α, which is non-constant for diagonal elements D_{kk}, and the reasoning from Section II-D no longer works.

Overall, **Non-identity** regularization proved to be the best, with the optimal parameter values for the test case of half-precision ($\varepsilon = 2^{-10}$), shown in Table III. **Scaled** L always improves the performance, but **Scaled** R_{UU} does not. The reason for that may lie in the fact that if H is aligned with interference subspace $V_{\text{sub}} = QR$, then $(I - QQ^H)H = 0$, so there is no channel outside of interference directions and the total noise power in the channel is proportional to interference power and not σ^2. This is indeed what we observe: there is about 15-20dB loss, when we add more high-power interference sources (in Fig. 3 and Fig. 4 compared to Fig. 2), meaning the noise power in the channel is increased by about 15-20dB, which can only mean that the channel is aligned with the interference. On the other hand, when there is only one interference source (Fig. 1, 2), increasing the interference power by 10dB does not lead to a 10dB loss in performance. Fig. 1 also shows that optimal regularization (eq. (8)) is preferred to stable regularization (eq. (9)), when the condition number of R_{UU} is low.

V. Conclusion

Simulations with the Quadriga 2.0 propagation channel demonstrate the proposed **Non-identity** regularization outperforms other standard regularization alternatives for the matrix inversion problem over half-precision Cholesky decomposition in Massive MIMO detector. The cost of regularization is negligible compared to the overall detector complexity.

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